Cut-and-Project Worldlines,
Geometric Gravitational Basins,
and the Emergence of Parabolic Visible Motion

Abstract

We develop a cut-and-project framework in which free affine motion in a higher-dimensional ambient space is observed as accelerated motion in a lower-dimensional visible sector. First, we prove a no-go statement: a free affine worldline remains affine under any fixed linear projection, so a visible parabola cannot arise from constant linear projection alone. Second, we show that a nonlinear observation map induces effective visible acceleration entirely through its Hessian. A minimal quadratic hidden-coordinate readout yields the exact ballistic law

$$y(t) = y_0 + v_{y,\mathrm{eff}}t - \frac{1}{2}g_{\mathrm{eff}}t^2, \qquad g_{\mathrm{eff}} = \kappa V_W^2.$$

Third, we formalize the notion of a geometric gravitational basin: the parabola is not the basin itself, but the local second-order visible trace of motion in such a basin. We define a basin operator, derive the local parabola theorem for smooth basins, obtain an induced metric and geodesic formulation, and state the design equation for extending the framework from constant-acceleration parabolic motion to Kepler-like inverse-square visible dynamics. The resulting hierarchy is

$$\text{trajectory} \subset \text{field} \subset \text{geometry}.$$

The document includes a figure suite showing affine-to-parabolic shadowing, parameter-space gravity maps, local-versus-global basin structure, and the role of Hessian order in generating observed acceleration.

Contents

1 Introduction

The guiding idea is that an observed lower-dimensional trajectory need not be fundamental. A visible worldline may be the shadow, cut, or nonlinear readout of a simpler ambient trajectory in a richer space. In this picture, what appears downstairs as force or acceleration can arise from curvature of the observation geometry rather than from a primitive lower-dimensional force law.

The framework has three layers:

1. **Ambient motion:** a worldline $X^A(\tau)$ in a higher-dimensional space.
2. **Observation law:** a map $x^\mu = f^\mu(X)$ producing visible coordinates.
3. **Effective visible dynamics:** the observed acceleration induced by the curvature of the observation map.

The central conceptual claim is:

> The parabola is not the basin itself. It is the local visible worldline generated by motion through a geometric gravitational basin.

Equivalently,

$$\text{parabolic arc} = \text{second-order visible trace of basin geometry.}$$

2 Ambient Free Motion and the Projection No-Go Result

Let the ambient worldline live in a five-dimensional space with coordinates

$$X^A(\tau) = (T(\tau), X(\tau), Y(\tau), Z(\tau), W(\tau)), \qquad A = 0, 1, 2, 3, 4.$$

Assume ambient free motion:

$$\frac{\mathrm{d}^2 X^A}{\mathrm{d}\tau^2} = 0. \tag{1}$$

Its general solution is affine,

$$X^A(\tau) = X_0^A + U^A \tau, \tag{2}$$

so in components,

$$T = T_0 + U_T \tau, \quad X = X_0 + U_X \tau, \quad Y = Y_0 + U_Y \tau, \quad Z = Z_0 + U_Z \tau, \quad W = W_0 + U_W \tau. \tag{3}$$

Suppose the visible coordinates are defined by a fixed linear projection,

$$x^\mu = P^\mu{}_A X^A, \qquad \mu = 0, 1, 2, 3, \tag{4}$$

with constant coefficients $P^\mu{}_A$.

Proposition 1 (Linear projection no-go)**.** *If $X^A(\tau)$ is affine in τ, then $x^\mu(\tau) = P^\mu{}_A X^A(\tau)$ is affine in τ. In particular, a visible parabola cannot arise from an affine ambient worldline under a fixed linear projection alone.*

Proof. Substituting (2) into (4) gives

$$x^\mu(\tau) = P^\mu{}_A X_0^A + P^\mu{}_A U^A \tau,$$

which is affine in τ. Hence

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2} = 0,$$

so no quadratic dependence on τ can appear. □

Remark 1. This result fixes the architecture of the theory. If a visible parabola is to arise from ambient free motion, then the observation mechanism must be nonlinear, curved, constrained, or frame-dependent.

3 Nonlinear Observation Maps and the Basin Operator

Let the visible coordinates be defined by a C^2 observation map

$$x^\mu = f^\mu(X^A). \tag{5}$$

Along the ambient affine worldline, the first derivative is

$$\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \frac{\partial f^\mu}{\partial X^A} \frac{\mathrm{d}X^A}{\mathrm{d}\tau} = \partial_A f^\mu \, U^A, \tag{6}$$

and the second derivative is

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2} = \partial_A \partial_B f^\mu \, U^A U^B + \partial_A f^\mu \frac{\mathrm{d}^2 X^A}{\mathrm{d}\tau^2}. \tag{7}$$

Using (1), the second term vanishes, so

$$\boxed{\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2} = \partial_A \partial_B f^\mu \, U^A U^B.} \tag{8}$$

Theorem 1 (Observation-curvature theorem)**.** *For a free affine ambient worldline and a C^2 observation map $x^\mu = f^\mu(X)$, the full visible acceleration is the directional Hessian of the observation map along the ambient velocity:*

$$a^\mu_{\text{obs}} = \partial_A \partial_B f^\mu \, U^A U^B.$$

Definition 1 (Basin operator)**.** *For each visible component μ, define the basin operator*

$$\boxed{\mathcal{B}^\mu_f[U,U](X) := \partial_A \partial_B f^\mu(X) \, U^A U^B.} \tag{9}$$

Then the visible dynamics are

$$\boxed{\ddot{x}^\mu = \mathcal{B}^\mu_f[U,U](X(\tau)).} \tag{10}$$

Remark 2. The parabola is not the primitive object. The Hessian structure of the observation map, encoded by $\mathcal{B}_f$, is the primitive object. The visible curve is the basin trace sampled by ambient motion.

4 Minimal Quadratic Hidden-Coordinate Model

Choose visible time and horizontal coordinate as ordinary readouts,

$$t = T, \qquad x = X, \qquad z = Z, \tag{11}$$

and define the visible height by

$$\boxed{y = Y - \frac{\kappa}{2} W^2,} \tag{12}$$

where κ is a constant observation-curvature parameter.

Fix the clock gauge $T = \tau = t$. Then the ambient affine solution becomes

$$X = X_0 + V_X t, \qquad Y = Y_0 + V_Y t, \qquad W = W_0 + V_W t. \tag{13}$$

Substituting into (12) gives

$$y(t) = Y_0 + V_Y t - \frac{\kappa}{2}(W_0 + V_W t)^2 \tag{14}$$

$$= \left(Y_0 - \frac{\kappa}{2} W_0^2\right) + (V_Y - \kappa W_0 V_W)\, t - \frac{\kappa}{2} V_W^2 t^2. \tag{15}$$

Define

$$y_0 := Y_0 - \frac{\kappa}{2} W_0^2, \qquad v_{y,\text{eff}} := V_Y - \kappa W_0 V_W, \qquad g_{\text{eff}} := \kappa V_W^2. \tag{16}$$

Then

$$\boxed{y(t) = y_0 + v_{y,\text{eff}} t - \frac{1}{2} g_{\text{eff}} t^2.} \tag{17}$$

Likewise,

$$x(t) = x_0 + v_x t, \qquad x_0 := X_0, \quad v_x := V_X. \tag{18}$$

Eliminating t yields

$$\boxed{y(x) = y_0 + \frac{v_{y,\text{eff}}}{v_x}(x - x_0) - \frac{g_{\text{eff}}}{2 v_x^2}(x - x_0)^2,} \tag{19}$$

which is a parabola.

Corollary 1 (Effective gravity)**.** *The visible vertical acceleration is*

$$\boxed{\ddot{y} = -g_{\text{eff}} = -\kappa V_W^2.} \tag{20}$$

Remark 3. This realizes the exact law

$$\text{constant hidden curvature} \Longrightarrow \text{uniform visible gravity} \Longrightarrow \text{parabola.}$$

5 Geometric Gravitational Basins

We now formalize the distinction between the basin and the curve traced inside it.

Definition 2 (Geometric gravitational basin)**.** *A geometric gravitational basin is any geometric structure - specified equivalently by a nonlinear observation map, an effective visible potential, or an induced metric/geodesic law - whose local curvature induces directed visible descent.*

Definition 3 (Visible basin field)**.** *Given an observation map $f^\mu(X)$ and ambient velocity U^A, define the visible effective field*

$$\mathcal{A}^\mu_f(X;U) := \mathcal{B}^\mu_f[U,U](X) = \partial_A\partial_B f^\mu(X)U^A U^B. \tag{21}$$

Definition 4 (Basin trace)**.** *A basin trace is a visible worldline generated by motion through a geometric gravitational basin.*

Proposition 2 (Parabola as basin trace)**.** *In the quadratic hidden-coordinate model, the visible parabola*

$$y(t) = y_0 + v_{y,\text{eff}}\,t - \frac{1}{2}g_{\text{eff}}t^2$$

is a basin trace, whereas the basin itself is the structure encoded by the readout

$$f^y(X,Y,Z,W) = Y - \frac{\kappa}{2}W^2$$

or equivalently by the induced metric derived below.

Remark 4. This is the precise statement of the intuition:

$$\text{parabola} \neq \text{basin}, \qquad \text{parabola} = \text{local visible worldline in a basin.}$$

6 Local Basin Expansion and the Local Parabola Theorem

Let $\Phi(\boldsymbol{r})$ be a smooth visible scalar potential on $\mathbb{R}^n$, with visible position $\boldsymbol{r} = (r^1,\ldots,r^n)$. Around a point $\boldsymbol{r}_0$, the Taylor expansion is

$$\Phi(\boldsymbol{r}) = \Phi(\boldsymbol{r}_0) + \partial_i\Phi(\boldsymbol{r}_0)\,\delta r^i + \frac{1}{2}\partial_i\partial_j\Phi(\boldsymbol{r}_0)\,\delta r^i\delta r^j + O(\|\delta\boldsymbol{r}\|^3), \tag{22}$$

where $\delta r^i = r^i - r^i_0$. The force law is

$$\ddot{r}^i = -\partial_i\Phi(\boldsymbol{r}). \tag{23}$$

If the Hessian term varies slowly on a local patch, then

$$\ddot{r}^i \approx -g^i, \qquad g^i := \partial_i\Phi(\boldsymbol{r}_0), \tag{24}$$

so the local motion is constant-acceleration motion.

Theorem 2 (Local parabola theorem)**.** *Let $\Phi(\boldsymbol{r})$ be a C^2 visible potential. In any sufficiently small neighborhood where $\nabla\Phi$ is approximately constant, the corresponding visible trajectory is, to second order in time, parabolic.*

Proof. In such a neighborhood,

$$\ddot{\boldsymbol{r}} \approx -\boldsymbol{g},$$

with constant $\boldsymbol{g}$. Hence

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t - \frac{1}{2}\boldsymbol{g}t^2.$$

Projecting onto any visible two-dimensional plane with one linear and one accelerated coordinate yields a parabola. □

Remark 5. This justifies the phrase

$$\boxed{\text{parabola} = \text{second-order visible grammar of a basin.}}$$

The parabola is the first nontrivial curvature signature seen by the visible observer.

7 Path, Field, and Geometry Hierarchy

The framework organizes naturally into three nested levels:

$$\boxed{\text{trajectory} \subset \text{field} \subset \text{geometry.}} \tag{25}$$

Level	Mathematical form	Interpretation
Trajectory	$y(x) = ax^2 + bx + c$	A specific visible path
Field	$\ddot{\boldsymbol{r}} = -\nabla\Phi$ or $\ddot{x}^\mu = \mathcal{A}^\mu_f$	Local descent law inside a basin
Geometry	$x^\mu = f^\mu(X)$ or $g_{ab}(q)$	Generative structure from which the field and trajectory arise

8 Potential, Observation Map, and Metric as Three Faces of One Basin

8.1 Observation-map form

The primary construction is

$$x^\mu = f^\mu(X^A), \qquad \ddot{x}^\mu = \partial_A\partial_B f^\mu\, U^A U^B.$$

8.2 Potential form

Suppose the visible spatial sector is x^i, $i = 1, \ldots, n$, and the effective field depends on visible position alone:

$$\ddot{x}^i = F^i(\boldsymbol{x}). \tag{26}$$

Definition 5 (Conservative visible basin)**.** *The visible field is conservative if there exists a scalar potential $\Phi(\boldsymbol{x})$ such that*

$$F^i(\boldsymbol{x}) = -\partial_i\Phi(\boldsymbol{x}). \tag{27}$$

Proposition 3 (Potential realization criterion)**.** *If the visible field is curl-free on a simply connected visible domain,*

$$\partial_i F_j - \partial_j F_i = 0, \tag{28}$$

then there exists a local scalar basin potential Φ *with* $F^i = -\partial_i \Phi$*.*

8.3 Metric/geodesic form

Let q^a denote coordinates on an effective manifold with metric $g_{ab}(q)$. Motion is governed by the geodesic Lagrangian

$$\mathcal{L} = \frac{1}{2} g_{ab}(q)\dot{q}^a \dot{q}^b, \tag{29}$$

and therefore by the geodesic equation

$$\frac{\mathrm{d}^2 q^a}{\mathrm{d}\lambda^2} + \Gamma^a{}_{bc} \frac{\mathrm{d}q^b}{\mathrm{d}\lambda} \frac{\mathrm{d}q^c}{\mathrm{d}\lambda} = 0. \tag{30}$$

Hence

$$\text{observation curvature} \longleftrightarrow \text{effective field} \longleftrightarrow \text{geodesic basin geometry.}$$

9 Induced Metric for the Quadratic Basin Model

Take the ambient flat metric

$$\eta = -\mathrm{d}T^2 + \mathrm{d}X^2 + \mathrm{d}Y^2 + \mathrm{d}Z^2 + \mathrm{d}W^2. \tag{31}$$

Introduce new coordinates (t, x, y, z, w) by

$$T = t, \qquad X = x, \qquad Y = y + \frac{\kappa}{2} w^2, \qquad Z = z, \qquad W = w. \tag{32}$$

Then

$$\mathrm{d}Y = \mathrm{d}y + \kappa w \, \mathrm{d}w.$$

Substituting into (31) gives

$$\begin{aligned} \mathrm{d}s^2 &= -\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}z^2 + (\mathrm{d}y + \kappa w \, \mathrm{d}w)^2 + \mathrm{d}w^2 && (33)\\ &= -\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 + 2\kappa w \, \mathrm{d}y \, \mathrm{d}w + (1 + \kappa^2 w^2)\mathrm{d}w^2. && (34) \end{aligned}$$

Thus the pulled-back metric is

$$\boxed{g = -\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2 + 2\kappa w \, \mathrm{d}y \, \mathrm{d}w + (1 + \kappa^2 w^2)\mathrm{d}w^2.} \tag{35}$$

The corresponding geodesic Lagrangian is

$$\mathcal{L} = \frac{1}{2} \left(-\dot{t}^2 + \dot{x}^2 + \dot{y}^2 + \dot{z}^2 + 2\kappa w \, \dot{y} \, \dot{w} + (1 + \kappa^2 w^2)\dot{w}^2 \right). \tag{36}$$

The momentum conjugate to y is

$$p_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = \dot{y} + \kappa w \dot{w}. \tag{37}$$

Since $\mathcal{L}$ does not depend explicitly on y, p_y is conserved:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} (\dot{y} + \kappa w \dot{w}) = 0. \tag{38}$$

Hence

$$\ddot{y} + \kappa(\dot{w}^2 + w\ddot{w}) = 0. \tag{39}$$

If the hidden-sector motion is affine,

$$w(\lambda) = w_0 + \nu\lambda, \qquad \dot{w} = \nu, \qquad \ddot{w} = 0, \tag{40}$$

then

$$\boxed{\ddot{y} = -\kappa\nu^2,} \tag{41}$$

which reproduces the same effective visible gravity as the observation-map derivation.

10 Local-to-Global Basin Hierarchy

A local basin patch generates parabolic traces, while a global central basin generates the full conic family.

10.1 Local basin patch

If the visible field is approximately constant over a local patch,

$$\ddot{\boldsymbol{r}} \approx -\boldsymbol{g},$$

then the trajectory is locally quadratic in time and parabolic after eliminating the linear coordinate.

10.2 Global central basin

If the visible potential is radial,

$$\Phi(r) = -\frac{\mu}{r}, \qquad r = \|\boldsymbol{r}\|, \tag{42}$$

then the visible equation of motion is

$$\ddot{\boldsymbol{r}} = -\mu\frac{\boldsymbol{r}}{r^3}, \tag{43}$$

whose orbits are ellipses, parabolas, or hyperbolas. Therefore,

$$\boxed{\text{local basin patch} \Rightarrow \text{parabolic trace}, \qquad \text{global central basin} \Rightarrow \text{conic family}.}$$

11 Kepler Design Equation from Ambient Free Motion

Let the visible spatial position be

$$\boldsymbol{r}(\tau) = \boldsymbol{f}(X^A(\tau)) \in \mathbb{R}^3, \tag{44}$$

with ambient affine motion $X^A(\tau) = X_0^A + U^A\tau$. Then

$$\ddot{r}^i = (U \cdot \nabla_X)^2 f^i(X), \qquad U \cdot \nabla_X := U^A\partial_A. \tag{45}$$

To reproduce Kepler-like visible motion,

$$\ddot{\boldsymbol{r}} = -\mu\frac{\boldsymbol{r}}{\|\boldsymbol{r}\|^3}, \tag{46}$$

the observation map must satisfy

$$\boxed{(U \cdot \nabla_X)^2 f^i(X) = -\mu\frac{f^i(X)}{\|\boldsymbol{f}(X)\|^3}, \qquad i = 1, 2, 3.} \tag{47}$$

Definition 6 (Kepler design equation)**.** *Equation (47) is the observation-map design equation for inducing inverse-square visible dynamics from ambient free motion.*

12 Basin Tensor Language

It is useful to isolate the ambient Hessian as a tensorial object.

Definition 7 (Basin tensor)**.** *For each visible component μ, define*

$$H^{\mu}{}_{AB}(X) := \partial_A \partial_B f^{\mu}(X). \tag{48}$$

Then

$$\ddot{x}^{\mu} = H^{\mu}{}_{AB} U^A U^B. \tag{49}$$

In the quadratic model,

$$H^{y}{}_{WW} = -\kappa, \qquad \text{all other relevant visible Hessian components vanish,} \tag{50}$$

so

$$\ddot{y} = H^{y}{}_{WW} V_W^2 = -\kappa V_W^2.$$

Remark 6. This makes the generative structure transparent:

$$\text{visible force} = \text{basin tensor} \times \text{ambient motion} \times \text{ambient motion}.$$

13 Figure Suite and Scientific Interpretation

The following figures are generated directly from the quadratic-basin and inverse-square models used in the note. They are not decorative; each figure corresponds to a specific structural statement in the theory.

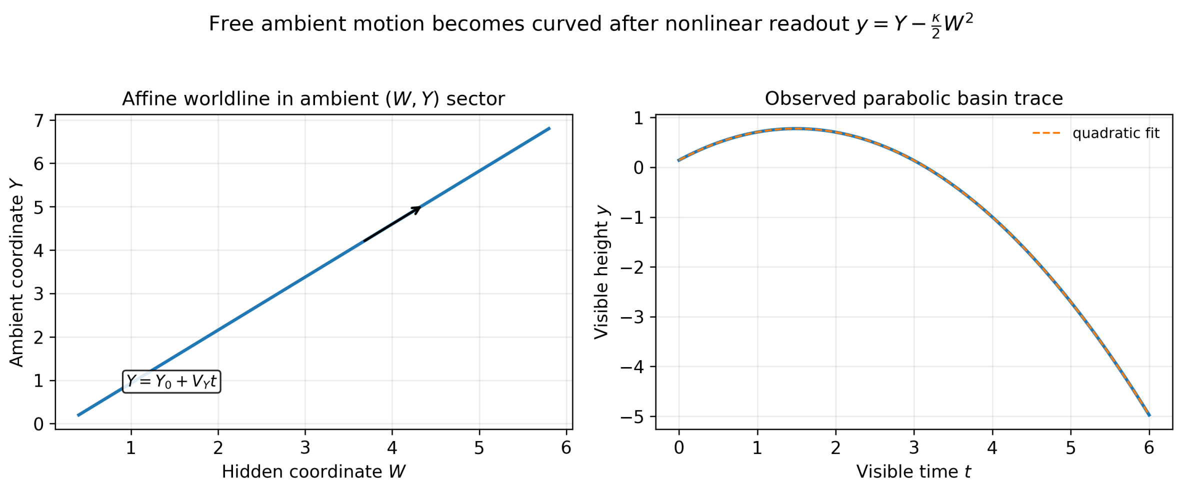

Figure 1: **Affine ambient motion versus observed parabolic basin trace.** Left: an affine worldline in the ambient (W, Y) sector. Right: the same motion observed through the nonlinear readout $y = Y - (\kappa/2)W^2$, producing a quadratic visible trajectory. This is the minimal demonstration that free motion upstairs can appear accelerated downstairs.

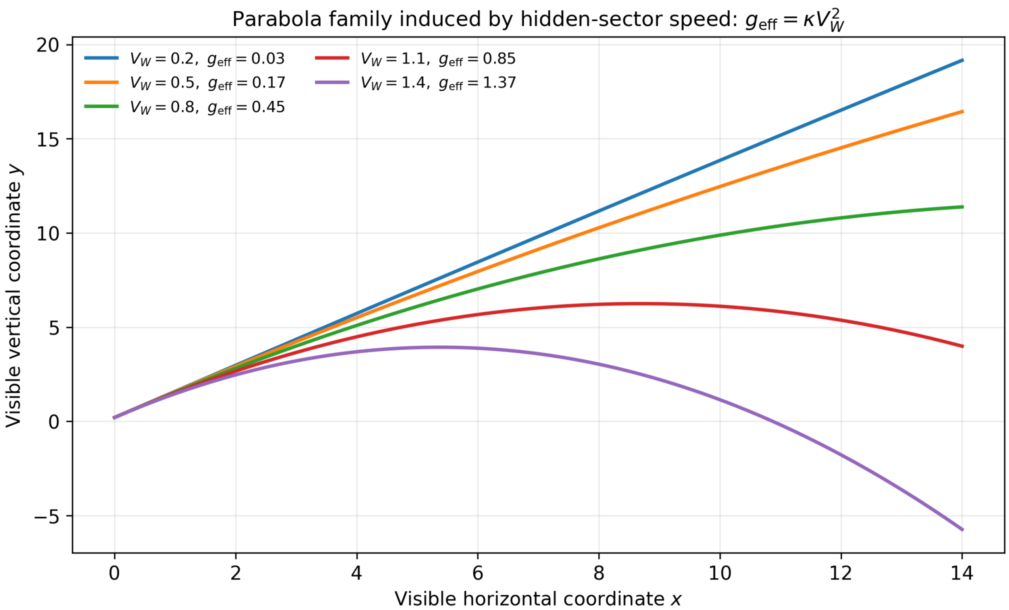

Figure 2: **Parabola family controlled by hidden-sector velocity.** The visible gravity obeys $g_{\mathrm{eff}} = \kappa V_W^2$, so increasing hidden-sector speed steepens the visible descent while leaving the ambient motion affine. This figure makes the parameter dependence explicit and suggests a direct numerical test: fit the visible curvature and verify quadratic scaling with V_W.

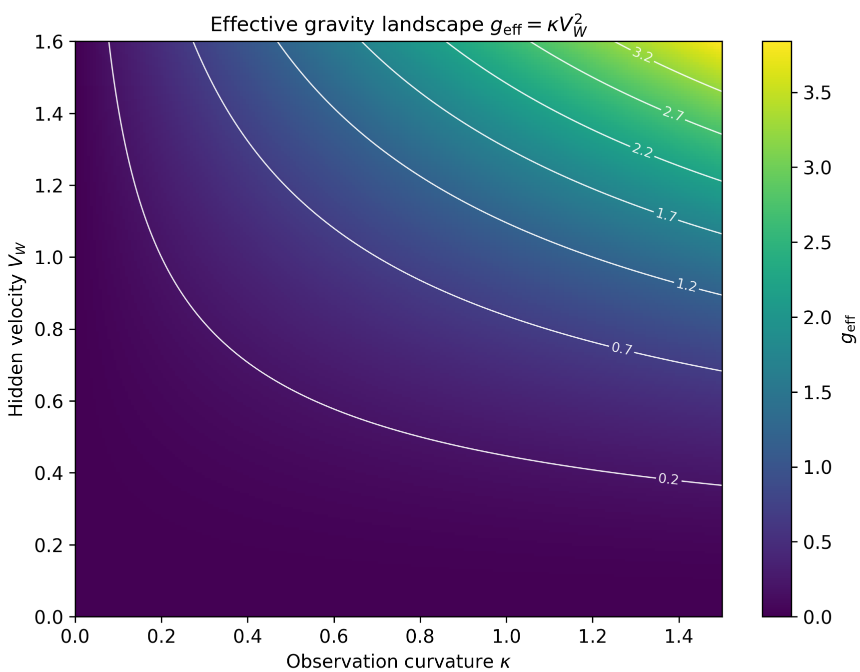

Figure 3: **Effective gravity landscape.** The parameter surface $g_{\mathrm{eff}} = \kappa V_W^2$ is shown over observation curvature κ and hidden velocity V_W. The level sets are quadratic in V_W and linear in κ, directly reflecting the basin-tensor contraction $\ddot{y} = H^y{}_{WW} V_W^2$.

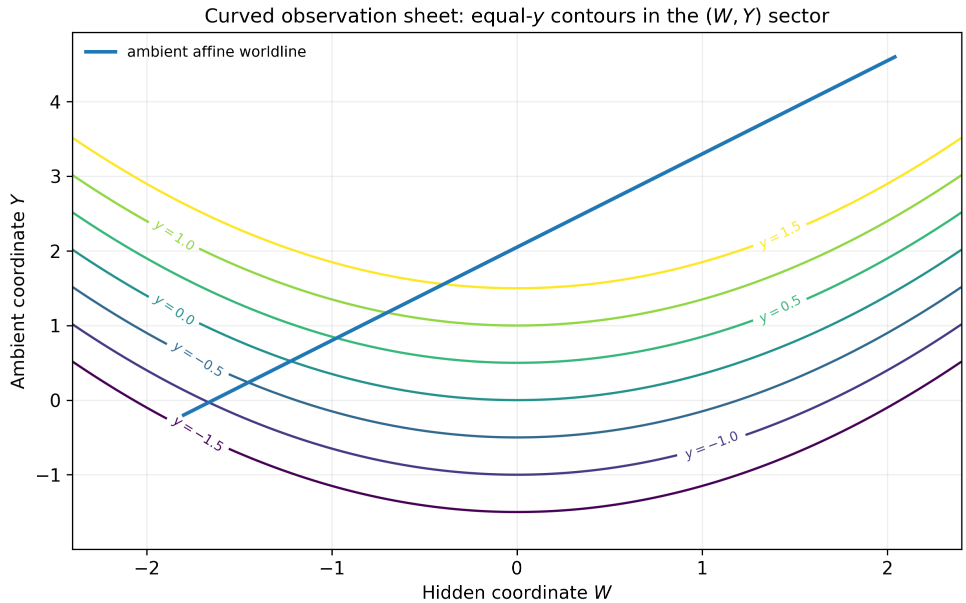

Figure 4: **Curved observation sheet in the (W, Y) sector.** Contours of constant visible height y define a curved observation sheet. The ambient affine worldline intersects these contours in a way that generates the visible parabolic basin trace. This figure makes the geometry concrete: the basin is encoded in the observation sheet, while the parabola is the resulting trace.

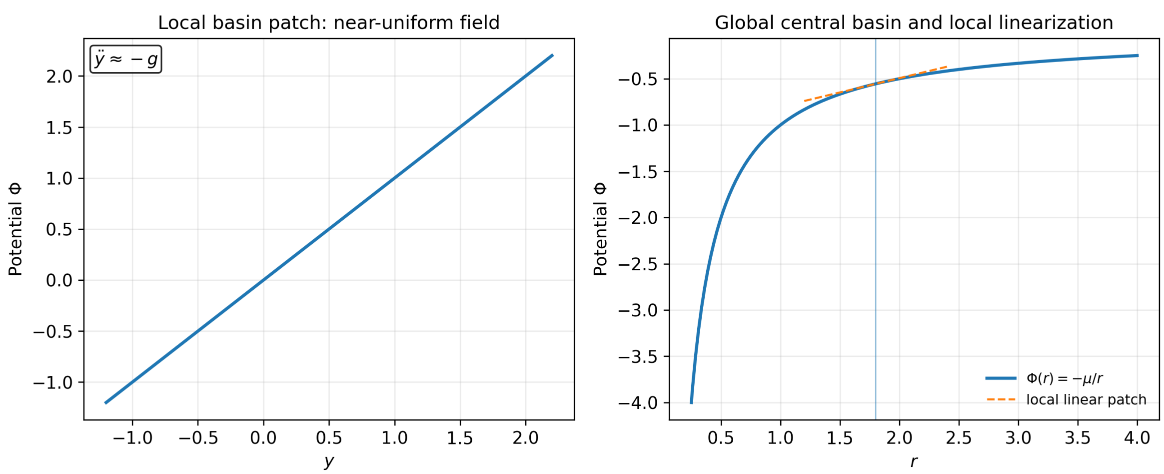

Figure 5: **Local and global basin structure.** Left: a local basin patch with approximately constant gradient, producing local constant-acceleration motion and therefore a parabola. Right: a global central basin $\Phi(r) = -\mu/r$ together with a local linearization around a reference radius. This is the precise local-to-global hierarchy of the framework.

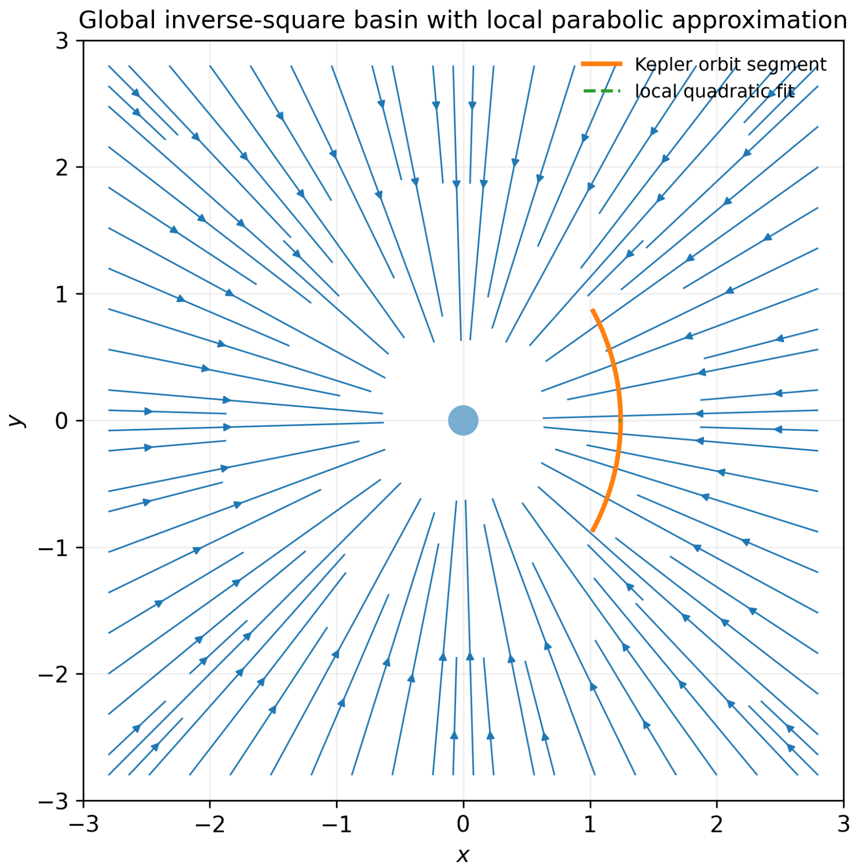

Figure 6: **Global inverse-square basin with a local quadratic fit.** The background streamlines represent the central inverse-square field. The orbit segment is globally Keplerian, but over a sufficiently small patch it is well approximated by a quadratic visible trace. This is the global conic family reducing locally to the parabolic grammar of a basin patch.

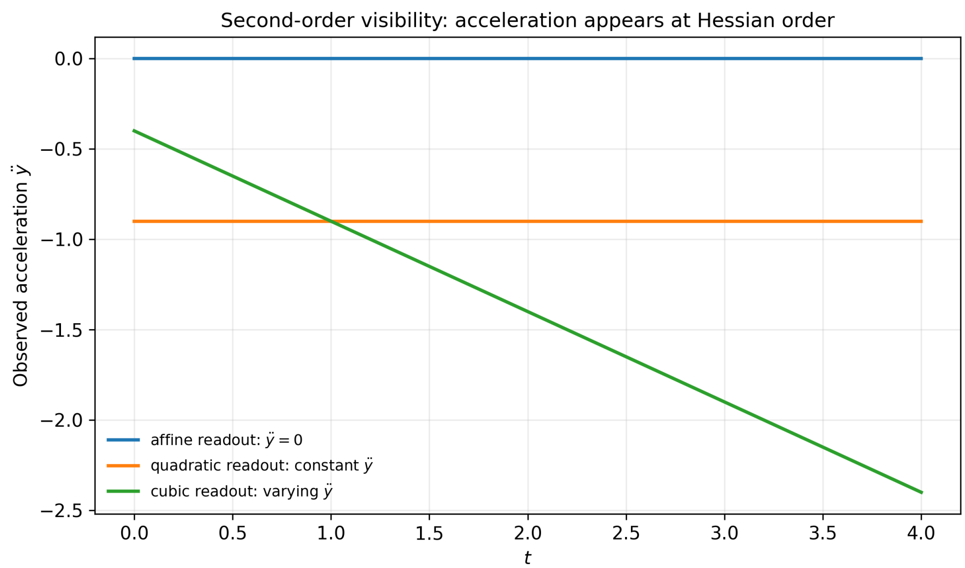

Figure 7: **Second-order visibility.** An affine readout has zero Hessian and therefore zero observed acceleration. A quadratic readout produces constant observed acceleration, while a cubic readout produces varying acceleration. The first nontrivial acceleration appears at Hessian order, which is why the parabola is the natural second-order visible signature of a basin.

14 Testable Consequences and Numerical Program

The framework makes several crisp, testable claims.

1. **Quadratic scaling law:**
$$g_{\text{eff}} = \kappa V_W^2.$$
Holding κ fixed and varying V_W should yield quadratic scaling of the fitted visible curvature.

2. **Readout-order dependence:** affine readouts produce zero visible acceleration; quadratic readouts produce constant visible acceleration; higher-order readouts produce time-varying visible acceleration.

3. **Local-to-global continuity:** local parabolic fits to a global inverse-square orbit should improve as the fitting window shrinks.

4. **Basin-tensor inference:** in numerical simulations with known observation map f^μ, the measured visible acceleration should match $H^\mu{}_{AB}U^AU^B$.

A natural computational workflow is:

1. sample affine ambient trajectories,
2. define candidate observation maps,
3. compute the basin tensor $H^\mu{}_{AB}$,
4. generate visible trajectories,
5. fit local parabolas and compare to analytic g_{eff},
6. search for maps approximately satisfying the Kepler design equation (47).

15 Master Summary

The complete framework can be compressed into the following chain:

$$
\begin{aligned}
\text{Ambient worldline:} \quad & X^A(\tau) = X_0^A + U^A\tau, \qquad \ddot{X}^A = 0, \\
\text{Observation law:} \quad & x^\mu = f^\mu(X), \\
\text{Visible acceleration:} \quad & \ddot{x}^\mu = \partial_A\partial_B f^\mu \, U^A U^B, \\
\text{Quadratic basin model:} \quad & y = Y - \frac{\kappa}{2}W^2, \\
\text{Effective gravity:} \quad & g_{\text{eff}} = \kappa V_W^2, \\
\text{Visible parabola:} \quad & y(t) = y_0 + v_{y,\text{eff}}t - \frac{1}{2}g_{\text{eff}}t^2, \\
\text{Geometric interpretation:} \quad & \text{parabola} = \text{local basin trace}, \\
\text{Global extension:} \quad & (U \cdot \nabla_X)^2 f^i(X) = -\mu \, f^i(X)/\left\|\boldsymbol{f}(X)\right\|^3 .
\end{aligned}
$$

16 Conclusion

We have formalized a coherent cut-and-project mechanics in which free higher-dimensional motion becomes accelerated visible motion through the curvature of the observation law. The framework supports the following compact statement:

> A parabola is the second-order visible shadow of motion through a geometric gravitational basin generated by deeper geometry.

The basin is primary; the visible curve is secondary. Local basin patches produce parabolic traces. Global central basins produce the conic family. Observation-map, potential, and metric descriptions are not competing accounts but three faces of the same generative structure.

The next rigorous step is numerical and constructive: search for observation maps whose basin tensors reproduce approximate inverse-square visible dynamics over large regions, not merely local parabolic behavior.

$$\boxed{\text{The gravitational basin is primary; the parabola is only its local projected worldline.}}$$